

AI Unit 2

4 Marks Questions

1. Define an Intelligent Agent

1. An intelligent agent is anything that can perceive its environment through sensors and act upon it using actuators. It observes what is happening around it and responds accordingly.
2. The information the agent receives at a particular moment is called a percept. All the percepts collected over time form a percept sequence, which helps the agent understand the situation.
3. The behavior of the agent is described by an agent function. This function maps the percept sequence to an action.
4. The actual implementation of the agent function in software or hardware is called the agent program. It runs on a computer or machine.
5. An intelligent agent aims to perform actions that help it achieve the best possible outcome according to a defined performance measure. Its intelligence depends on how effectively it achieves its goal.

2. What is a Rational Agent?

1. A rational agent is one that always chooses the action that maximizes its expected performance measure. It tries to do the “right thing” based on the situation.
2. It makes decisions using its percept sequence, built-in knowledge, available actions, and the performance measure defined for the task.
3. Rationality does not mean the agent knows everything about the future. It only uses the information it currently has and selects the best possible action.
4. A rational agent may sometimes perform actions to gather more information if that improves future decisions.
5. The performance measure must reflect the real goal. If it is poorly defined, the agent may behave in unintended ways, even though it is technically rational.

3. Explain the Concept of Agent Environment with an Example

1. The agent environment includes everything that surrounds the agent and affects its actions. It is the world in which the agent operates.
2. Before designing an agent, we define the task environment using PEAS (Performance measure, Environment, Actuators, Sensors). This helps clearly describe the problem.
3. For example, consider an automated taxi driver. Its performance measure includes reaching the destination safely, minimizing fuel use, and ensuring passenger comfort.
4. The taxi's environment includes roads, traffic, pedestrians, weather, and passengers. It must deal with dynamic and changing conditions.
5. The taxi uses sensors like cameras, GPS, and speedometers to collect information. It uses actuators like steering, brakes, and accelerator to take actions.

4. List and Explain Any Two Types of Agents

(a) Fully Observable Agent

1. In a fully observable environment, the agent can access the complete state of the environment at every moment. No important information is hidden.
2. The agent does not need to remember past states because the current percept provides all necessary information.
3. Decision-making is simpler since there is no uncertainty about the environment. The agent can directly choose the best action.
4. An example is a chess-playing AI. The entire board is visible, so the agent always knows the exact state of the game.

(b) Partially Observable Agent

1. In a partially observable environment, the agent cannot see the complete state of the environment. Some information may be hidden or unclear.
2. The agent may need memory or reasoning to estimate missing information. It cannot rely only on the current percept.
3. This type of environment introduces uncertainty, making decision-making more complex.
4. An example is a vacuum cleaner that senses dirt only in its current location. It does not know the condition of other locations unless it moves there.

5. What is a Learning Agent?

1. A learning agent is an intelligent agent that improves its performance over time through experience. It learns from past percepts and actions.
2. Learning helps the agent adapt to changes in the environment. Without learning, the agent may fail when conditions change.
3. A learning agent updates its internal knowledge based on feedback. This allows it to correct mistakes and refine its behavior.
4. Learning increases the autonomy of the agent because it depends more on its own experience rather than only on built-in knowledge.
5. Such agents are more flexible, robust, and capable of performing well in complex and unpredictable environments.

6. Write Short Note on Structure of an Intelligent Agent

1. The structure of an intelligent agent consists of **sensors, actuators, agent function, and agent program**. Sensors help the agent receive information from the environment, while actuators help it perform actions.
2. The agent function maps the percept sequence to an action. It defines what action the agent should take for every possible situation.
3. The agent program is the real implementation of the agent function. It runs on a physical system such as a computer or robot.
4. The agent interacts continuously with the environment by sensing and acting. Its structure ensures proper decision-making to achieve goals effectively.

7. Define Problem-Solving Agent

1. A problem-solving agent is a type of intelligent agent that decides what to do by searching for a sequence of actions that leads to a goal. It focuses on achieving a clearly defined objective.
2. It first identifies the goal and then considers possible actions that can help reach that goal. The agent chooses the best sequence of actions.
3. Such an agent operates in a well-defined environment where states, actions, and goals are clearly specified.
4. It uses reasoning and search techniques to solve problems step by step. Chess-playing AI and route-finding systems are common examples of problem-solving agents.

8. Give Examples of Environments in Which AI Agents Operate

1. One example is an **automated taxi driving environment**. The taxi operates on roads with traffic, pedestrians, signals, and weather conditions.
2. A **chess-playing environment** is another example. The AI agent interacts with another player and makes moves based on the board state.
3. A **vacuum cleaner world** is a simple example environment. The agent senses dirt and moves between locations to clean.
4. A **robot working in a warehouse** is also an example. It navigates through shelves and objects to complete tasks.
5. These environments can be fully observable, partially observable, static, dynamic, deterministic, or stochastic depending on their nature.

9. What are the Components of a Problem in Problem Solving?

1. The **initial state** defines where the agent starts. It describes the current situation of the problem.
2. The **actions** specify the possible moves the agent can take. These actions change the state of the environment.
3. The **transition model** describes the result of performing an action in a state. It shows how one state changes to another.
4. The **goal test** checks whether the desired goal has been achieved. It helps the agent know when to stop.
5. The **path cost** measures the total cost of actions taken. The agent tries to find the solution with minimum cost.

10. Differentiate Between Deterministic and Stochastic Environments

1. In a **deterministic environment**, the next state is completely determined by the current state and the action taken. There is no uncertainty in the outcome.
2. In a deterministic environment, if the agent performs the same action in the same state, the result will always be the same. Chess is an example of a deterministic environment.
3. In a **stochastic environment**, the outcome of an action is uncertain. The same action may produce different results at different times.
4. Stochastic environments involve probabilities and randomness. Taxi driving is an example because traffic and other drivers' actions are unpredictable.
5. Deterministic environments are easier to handle, while stochastic environments require handling uncertainty and risk.

1. Explain the Concept of Rationality in Intelligent Agents

1. Rationality in intelligent agents means choosing the action that gives the best expected result. A rational agent always tries to maximize its performance measure.
2. The decision of a rational agent depends on four things: the percept sequence, built-in knowledge, available actions, and the performance measure. It selects the best possible action based on these factors.
3. Rationality does not mean the agent knows everything about the future. It simply means the agent makes the best decision using the information it currently has.
4. A rational agent may gather more information if it improves future decisions. Proper design of the performance measure is very important, otherwise the agent may behave in unintended ways.

2. Describe the Structure of an Intelligent Agent with a Neat Diagram

1. The structure of an intelligent agent includes **sensors, actuators, agent function, and agent program**. Sensors collect information from the environment, and actuators perform actions.
2. The agent function maps the percept sequence to an action. The agent program implements this function in real hardware or software.
3. The agent continuously interacts with the environment by sensing and acting. This cycle helps it achieve its goals.
4. The structure can be represented as:

Environment > Sensors > Agent Program > Actuators > Environment

5. This diagram shows how the agent receives input from the environment and sends actions back to it.

3. Explain Types of Agents in Detail with Examples

1. Fully Observable Agents - Fully observable agents operate in environments where the complete state of the environment is available through sensors. No important information is hidden, so the agent can make decisions using only the current percept.

These agents do not need to rely heavily on memory of past states. For example, a chess-playing AI can see the entire chessboard at all times and knows the exact position of every piece.

2. Partially Observable Agents - Partially observable agents work in environments where some information is hidden or uncertain. The sensors do not provide complete details about the environment.

These agents must use memory or reasoning to handle missing information. For example, a vacuum cleaner that senses dirt only in its current location does not know about other locations unless it moves there.

3. Single-Agent Environments - In single-agent environments, only one agent's actions affect the outcome. There are no other agents competing or influencing the results.

The agent only needs to consider fixed rules or physical conditions. For example, solving a crossword puzzle alone depends only on the individual's decisions.

4. Multiagent Environments - Multiagent environments contain more than one agent, and each agent may have its own goals. The actions of one agent can affect the success of others.

These environments can be competitive or cooperative. For example, chess is competitive because one player's win is the other's loss.

5. Deterministic and Stochastic Environments - In deterministic environments, the next state is completely determined by the current state and the action taken. The same action always produces the same result.

In stochastic environments, the outcome of an action is uncertain and may involve randomness. Taxi driving is stochastic because traffic and road conditions are unpredictable.

4. Describe a Learning Agent and Its Components

1. A learning agent is an agent that improves its performance over time by learning from experience. It adapts to changes in the environment.
2. The learning agent has four main components: the performance element, learning element, critic, and problem generator.
3. The performance element selects actions based on current knowledge. It directly interacts with the environment.
4. The learning element improves the performance element by updating knowledge. The critic provides feedback on how well the agent is performing.
5. The problem generator suggests new actions to explore, helping the agent gain new experiences.

5. Explain the Nature of Environments in AI with Examples

1. A **fully observable environment** is one where the agent can see the complete state. Chess is an example because the entire board is visible.
2. A **partially observable environment** is one where some information is hidden or uncertain. Poker is an example since players cannot see opponents' cards.
3. A **deterministic environment** is one where the outcome of actions is predictable. The same action always leads to the same result.
4. A **stochastic environment** involves randomness and probabilities. Taxi driving is stochastic because traffic conditions are unpredictable.
5. Understanding the nature of the environment helps in designing a suitable intelligent agent.

6. Explain the Concept of a Problem-Solving Agent

1. A problem-solving agent is a type of intelligent agent that decides what to do by searching for a sequence of actions that leads to a goal. It focuses mainly on achieving a clearly defined objective.
2. The agent first identifies the current state of the environment. Then it defines the goal state that it wants to reach.
3. It considers all possible actions that can change the current state into another state. The agent explores different paths to reach the goal.
4. Using search techniques, the agent selects the best sequence of actions. It usually tries to minimize cost or number of steps.
5. Examples of problem-solving agents include route-finding systems and puzzle-solving programs like the 8-puzzle.

7. Explain the Steps Involved in Formulating a Problem for AI Search

1. The first step is defining the **initial state**. This represents the starting condition of the agent.
2. The second step is defining the **actions** available to the agent. These actions describe what the agent can do to change the state.
3. The third step is specifying the **transition model**. It explains the result of performing a particular action in a given state.
4. The next step is defining the **goal test**. It checks whether the desired goal state has been reached.
5. Finally, the **path cost** is defined. It measures the total cost of actions taken, and the agent tries to find a solution with minimum cost.

8. Describe One Example Problem Solved Using Search (Route Finding)

1. Route finding is a common example of a problem solved using search. The agent's goal is to find the best path from a starting location to a destination.
2. The **initial state** is the starting city, and the **goal state** is the destination city. The agent must move through different intermediate cities.
3. The possible actions are traveling from one city to another using available roads. The **transition model** shows which city is reached after taking a road.
4. Each road may have a distance or cost, which is called the **path cost**. The agent tries to find the shortest or least costly path.
5. The agent uses search strategies to explore possible routes and selects the best path that satisfies the goal.

9. Compare Goal-Based Agents and Utility-Based Agents

1. Goal-based agents focus only on achieving a specific goal. They check whether a particular state satisfies the goal condition.
2. These agents do not measure how good a solution is, as long as it achieves the goal. Their decision-making is based mainly on reaching the desired state.
3. Utility-based agents evaluate different possible outcomes using a utility function. They choose actions that maximize overall benefit or satisfaction.
4. Utility-based agents can handle situations with multiple goals or trade-offs. They compare different results and select the most desirable one.
5. Goal-based agents are simpler, while utility-based agents are more flexible and powerful in complex environments.

10. Explain the Role of Percepts in an Agent's Decision-Making Process

1. A percept is the information an agent receives from the environment at a particular moment. It is collected through sensors such as cameras or input devices.
2. The agent stores percepts over time as a percept sequence. This sequence represents the complete history of what the agent has observed.
3. The agent function uses the percept sequence to decide which action to take. Therefore, percepts directly influence the agent's behavior.
4. If percepts are incomplete or incorrect, the agent may make wrong decisions. Accurate percepts help the agent act rationally and achieve better performance.
5. Without percepts, the agent cannot understand the environment, and effective decision-making would not be possible.

1. Define Intelligent Agents and Explain in Detail

Definition of Intelligent Agent

1. An intelligent agent is an entity that perceives its environment through sensors and acts upon that environment using actuators. It continuously interacts with the environment by sensing and acting.
2. The information received at a particular moment is called a percept. All percepts collected over time form a percept sequence, which helps the agent decide what action to take.

Agents and Environments

1. An agent operates within an environment, which includes everything that affects its actions. The environment provides inputs to the agent through sensors.
2. The agent responds using actuators to perform actions that influence the environment. For example, in a vacuum cleaner world, the agent senses dirt and moves or cleans accordingly.
3. The task environment can be described using PEAS (Performance, Environment, Actuators, Sensors). This helps clearly define the problem the agent must solve.

Concept of Rationality

1. A rational agent chooses the action that maximizes its expected performance measure. It tries to do the “right thing” based on available information.
2. Rationality depends on percept sequence, built-in knowledge, available actions, and performance measure. It does not mean the agent knows the future.
3. Proper design of the performance measure is important. Otherwise, the agent may behave in unintended ways.

Nature of Environment

1. Environments can be fully observable or partially observable. In fully observable environments, the agent can see the complete state.
2. They can also be deterministic or stochastic. Deterministic environments have predictable outcomes, while stochastic environments involve uncertainty.
3. Environments may be static or dynamic, and single-agent or multiagent. Understanding the nature of the environment helps design better agents.

Structure of Agents

1. The structure of an agent includes sensors, actuators, agent function, and agent program. Sensors collect information, and actuators perform actions.
2. The agent function maps percept sequences to actions. The agent program implements this function in real systems.
3. The agent continuously senses and acts in a loop to achieve its goal efficiently.

2. Explain the types of agents in detail:

1. Simple Reflex Agent

1. A simple reflex agent acts only based on the current percept. It does not store past information or consider future consequences.
2. It uses condition–action rules (if–then rules) to decide actions. When a condition is true, the corresponding action is performed.
3. It works effectively only in fully observable environments. For example, a simple vacuum cleaner that sucks dirt whenever it detects dirt.

2. Model-Based Agent

1. A model-based agent maintains an internal state to keep track of past percepts. This helps it handle partially observable environments.
2. It uses a model of how the world works to update its internal state. The model helps predict how actions affect the environment.
3. This type of agent makes better decisions than simple reflex agents. For example, a vacuum cleaner that remembers which locations are clean or dirty.

3. Goal-Based Agent

1. A goal-based agent acts to achieve a specific goal. It considers whether a state satisfies the goal condition.
2. It uses search and planning to find a sequence of actions that leads to the goal. It evaluates possible future states.
3. This agent is more flexible because it can choose different actions to achieve the same goal. For example, route-finding systems.

4. Utility-Based Agent

1. A utility-based agent not only tries to achieve a goal but also measures how desirable a state is. It uses a utility function.
2. It compares different possible outcomes and chooses the one with maximum utility. This helps in complex decision-making.
3. It can handle trade-offs between different goals. For example, choosing the fastest yet safest route in taxi driving.

5. Learning Agent

1. A learning agent improves its performance over time by learning from experience. It adapts to changes in the environment.
2. It has components like performance element, learning element, critic, and problem generator. These components help improve decisions.
3. Learning agents are more autonomous and flexible. They depend on experience rather than only built-in knowledge.

3. Explain the relationship between agents and their environments. Fully observable vs partially observable , Deterministic vs stochastic , Episodic vs sequential , Static vs dynamic environments

Relationship Between Agents and Their Environments

An agent always operates within an environment. The environment provides inputs to the agent through sensors, and the agent affects the environment through actions using actuators. The nature of the environment directly influences how the agent is designed and how it makes decisions.

1. Fully Observable vs Partially Observable

1. In a **fully observable environment**, the agent can access the complete state of the environment at every moment. It does not need to rely on memory because all necessary information is available. Example: Chess, where the entire board is visible.
2. In a **partially observable environment**, some information is hidden or uncertain. The agent must use memory or reasoning to make decisions. Example: A vacuum cleaner that senses dirt only in its current location.

2. Deterministic vs Stochastic

1. In a **deterministic environment**, the next state is completely determined by the current state and the action taken. The same action always produces the same result.
2. In a **stochastic environment**, the outcome of an action is uncertain or involves randomness. Taxi driving is an example because traffic and other drivers are unpredictable.

3. Episodic vs Sequential

1. In an **episodic environment**, each decision is independent of previous decisions. The current action does not affect future episodes. Example: Identifying defective products on an assembly line.
2. In a **sequential environment**, current decisions affect future states and decisions. Example: Chess or taxi driving, where each move influences future outcomes.

4. Static vs Dynamic Environments

1. In a **static environment**, the environment does not change while the agent is making a decision. The agent can take its time to decide. Example: Crossword puzzle.
2. In a **dynamic environment**, the environment changes continuously while the agent is deciding. The agent must respond quickly. Example: Taxi driving, where other vehicles keep moving.

4. Identify the type of agent suitable for the following tasks:

Chess playing , Self-driving car , Smart thermostat

1. Chess Playing – Goal-Based / Utility-Based Agent

1. A chess-playing system is mainly a goal-based agent because its primary objective is to achieve the goal of checkmating the opponent. It searches for a sequence of moves that leads from the current board state to a winning state.
2. It considers future consequences of its moves before acting. The agent evaluates possible game states and chooses moves that help reach the goal.
3. It is also a utility-based agent because it uses an evaluation (utility) function to measure how good a board position is. It selects moves that maximize its chances of winning.

2. Self-Driving Car – Utility-Based / Learning Agent

1. A self-driving car is a utility-based agent because it must balance multiple factors like safety, speed, fuel efficiency, and passenger comfort. It selects actions that maximize overall utility.
2. The environment is dynamic and stochastic, so the car must continuously evaluate different possible outcomes before making decisions.
3. It is also a learning agent because it improves performance over time. It learns from driving data and adapts to different traffic and road conditions.

3. Smart Thermostat – Simple Reflex / Model-Based Agent

1. A basic smart thermostat works as a simple reflex agent. It follows simple condition–action rules such as turning the heater on when the temperature drops.
2. It reacts only to the current temperature reading and does not consider past data. This makes it suitable for simple environments.
3. Advanced thermostats act as model-based agents because they store past temperature patterns. They use this internal model to adjust heating or cooling more efficiently